LedgerForge — Phase 3 Report

WorkforceOS: Connected System — Expenses, Payroll-to-Ledger, JWT RBAC, Tax Engine & Financial Statements

Candidate Name: Ancillin P G | **Track:** Python Developer
Phase: Phase 3: Connected System (Days 11 to 15 / Tasks LF-11 to LF-15) | **Git Tag:** phase-3
GitHub Repository: https://github.com/Ancilin/Foundation-Project-Setup-Employee-Data-Model

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1. Phase 3 Task Execution & Evidence Table (Days 11 to 15)

Day / Task	Objective & Scope	Status	Deliverables & Clickable Evidence Links
Day 11 LF-11	**Spend Money: Vendors & Expense Bills** 56 vendors, 1,105 bills, SUBMITTED -> APPROVED -> PAID lifecycle, separation of duties (approved_by != submitted_by), GST Input (1200) + AP (2001), duplicate detector.	DONE	• Service: app/services/expense_service.py • Routes: /api/v1/vendors, /api/v1/expenses • Spend Report: 18/18 months match expenses.csv (diff=0.00) • Tests: tests/test_expenses.py (3/3 passed)
Day 12 LF-12	**Connect Payroll to the Books (All 18 Months)** End-to-end payroll lifecycle (DRAFT -> PROCESSED -> APPROVED -> PAID), idempotency key (payroll:YYYY-MM:ZNT), single JE per run, Check 2 reconciliation, run reversal.	DONE	• Services: payroll_service.py, ledger_service.py • Check 2: 18/18 runs equal ledger (largest diff = 0.00) • Idempotency: duplicate approval returns same JE without double-posting • Tests: tests/test_payroll_to_ledger.py (3/3 passed)
Day 13 LF-13	**Lock the Doors: Logins, Roles & Permissions** 745 users with bcrypt hashes, POST /auth/login (30-min JWT via PyJWT), SECRET_KEY from env, reusable require_role(), employee self-access guard, deactivated-user 401 guard.	DONE	• Security: app/core/security.py, routes/auth.py • Matrix: GET /api/v1/auth/permissions (all routes protected) • Live Deactivation Check: active JWT rejected with 401 USER_DEACTIVATED • Tests: tests/test_auth_security.py (5/5 passed across 6 routes)
Day 14 LF-14	**Taxes & Statutory Reports: Configurable Rules** DB-driven tax_slabs, tax_parameters, statutory_rates; Admin-only slab mutation; single-month rerun without altering past months; monthly PF, ESI, and TDS reports.	DONE	• Service & Routes: app/services/tax_service.py, routes/tax.py • Rerun Proof: EMP-0006 (2026-09) TDS 51,690.73 -> 53,520.73 -> 51,690.73 • Statutory Reconciliation: PF, ESI & TDS diff = 0.00 across 18 months • Tests: tests/test_tax_and_statutory_reports.py (3/3 passed)
Day 15 LF-15	**Financial Story: Trial Balance, P&L; & Balance Sheet** Date-range Trial Balance, Profit & Loss, and Balance Sheet excluding parent group accounts (1000-5000); trial-balance CSV verifier (0 mismatches); Assets = Liabilities + Equity + Profit.	DONE	• Service & Routes: app/services/report_service.py, routes/reports.py • Trial Balance Verifier: Accounts compared: 40, mismatched: 0 • Official Match: Rev 1,260,930,052.00, Exp 1,036,246,464.42, Profit 224,683,587.58 • Tests: tests/test_financial_reports.py (3/3 passed)

2. Mini Challenge: Tracing One Invoice & One Payroll Run from Input to the Balance Sheet

End-to-End Accounting Architecture Flow (Operational Input -> Double-Entry Journal -> Financial Statements):
[1. Input Event] ---> [2. Sub-Ledger Service] ---> [3. Posted Journal Entry] ---> [4. Trial Balance] ---> [5. P&L & Balance Sheet]

• **Invoice Path (INV-00052):** Sent Invoice (16,64,744) -> JE-000007 (Dr 1100 AR / Cr 4004 Rev + Cr 2200 GST) -> Receipt PAY-00053 -> JE-000227 (Dr 1001 Bank / Cr 1100 AR) -> P&L Revenue (+14,10,800) -> BS Current Profit (+14,10,800), Bank (+16,64,744), GST Payable (+2,53,944).

• **Payroll Path (PR-2026-08):** 429 Attendance/Salary Rows -> Approved Run (payroll:2026-08:ZNT) -> JE-002373 (Dr 5001-5005 Cost 4,93,95,920.24 / Cr 2100-2140, 1300) -> Disbursement (Dr 2100 / Cr 1001 Bank) -> P&L Expense (+4,93,95,920.24) -> BS Current Profit (-4,93,95,920.24).

Trace A — Client Invoice INV-00052 (NorthStar Telecom Ltd):
1. Input & Tax: Service account 4004, Subtotal **INR 1,410,800.00** + 18% GST **INR 253,944.00** = Total **INR 1,664,744.00** (DRAFT posts nothing).
2. Accrual Journal (JE-000007 on 2025-04-01): Marking SENT posts **Dr 1100 AR 1,664,744.00**; **Cr 4004 Revenue 1,410,800.00**; **Cr 2200 GST Output 253,944.00**.
3. Receipt Journal (PAY-00053 / JE-000227 on 2025-05-26): RTGS receipt clears invoice to PAID and posts **Dr 1001 Bank 1,664,744.00** / **Cr 1100 AR 1,664,744.00** (net AR 1100 = 0.00).
4. Trial Balance -> P&L -> Balance Sheet: 4004 adds **+1,410,800.00** to P&L Revenue and BS current_period_profit; plus Liability 2200 (**+253,944.00**) = Asset 1001 Bank (**+1,664,744.00**).

Trace B — Payroll Run PR-2026-08 (429 Employees, Aug 2026):
1. Engine Input: Attendance + DB tax slabs yield Gross **48,685,007.97**, Deductions **5,511,689.61**, Net **43,173,318.36**, Employer PF/ESI **710,912.27** (Total Cost = **INR 49,395,920.24**).
2. Accrual Journal (JE-002373, key payroll:2026-08:ZNT): Debits Expense 5001 (48,492,646.56), 5002 (167,361.41), 5003 (25,000.00), 5004 (708,198.35), 5005 (2,713.92) = **49,395,920.24**; **Credits** 2100 Net Pay (43,173,318.36), 2110 PF (1,416,396.70), 2120 ESI (3,340.22), 2130 TDS (4,708,264.96), 2140 PT (82,600.00), 1300 Advances (12,000.00).
3. Disbursement & Balance Sheet: Payment debits 2100 / credits 1001 Bank (**43,173,318.36**). Leaf accounts 5001-5005 add **49,395,920.24** to P&L Expenses, reducing BS current_period_profit by **49,395,920.24**, matching the net change in Assets & Liabilities.

3. Proof It Works: Official Verifier Outputs & Automated Test Suite

A. Official Ledger Integrity Checks, Trial Balance Verifier & Headline Summary (verify_ledgerforge.py)

```
$ python scripts/verify_ledgerforge.py checks --data exports
PASS Every journal entry balances - debits-credits=0.0, unbalanced entries=0
PASS Payroll equals ledger, every month - largest monthly difference=0.00
PASS Receivables equal open invoices - ledger 1100=139,394,278.00, open (sent/overdue/part-paid) invoices=139,394,278.00
PASS Payroll employees exist - payroll rows with unknown employee=0
PASS Nobody is paid after leaving - payroll rows after exit_date=0
5/5 checks passed

$ python scripts/verify_ledgerforge.py trial-balance exports/trial_balance.csv --data data/csv
Accounts compared: 40, mismatched: 0

$ python scripts/verify_ledgerforge.py summary --data exports
Employees: 745 (active 441, exited 304) | Attendance rows: 150,155 | Payroll runs: 18, items: 7,340, employer cost:
861,586,392.42
Invoices: 743, total billed: 1,419,619,092.46 | Expenses: 1,105, total gross: 202,914,225.72
Journal entries: 2,520, lines: 8,255 | Revenue: 1,260,930,052.00 | Expenses: 1,036,246,464.42 | Profit: 224,683,587.58
```

B. Financial Statements Reconciliation (Official Summary vs API Reports)

Financial Report / Metric	API Endpoint	Official Target (INR)	LedgerForge API Output (INR)	Variance
Trial Balance Total Debits	GET /api/v1/reports/trial-balance	3,654,484,222.74	**3,654,484,222.74** (40 leaf accts)	**0.00**
Trial Balance Total Credits	GET /api/v1/reports/trial-balance	3,654,484,222.74	**3,654,484,222.74** (is_balanced=True)	**0.00**
P&L Total Revenue (4001-4010)	GET /api/v1/reports/profit-and-loss	1,260,930,052.00	**1,260,930,052.00**	**0.00**
P&L Total Expenses (5001-5112)	GET /api/v1/reports/profit-and-loss	1,036,246,464.42	**1,036,246,464.42**	**0.00**
P&L Net Profit (Revenue - Expenses)	GET /api/v1/reports/profit-and-loss	224,683,587.58	**224,683,587.58**	**0.00**
Balance Sheet: Total Assets	GET /api/v1/reports/balance-sheet	463,762,958.65	**463,762,958.65**	**0.00**
Balance Sheet: Liab + Equity + Profit	GET /api/v1/reports/balance-sheet	463,762,958.65	**77,499,607.07 + 161,579,764.00 + 224,683,587.58**	**0.00**

C. Automated Pytest Suite (113 Passed, 0 Failed)

```
============================= test session starts ==============================
platform win32 -- Python 3.12.10, pytest-8.4.2
collected 113 items

tests/test_expenses.py (3 tests: lifecycle->AP->Bank, self-approval 400, spend report + duplicates) ...... PASSED
tests/test_payroll_to_ledger.py (3 tests: Check 2 all 18 runs=0.00, idempotent approve, reversal) ....... PASSED
tests/test_auth_security.py (5 tests: 401/403 across 6 routes, payslip IDOR 403, deactivated token 401) . PASSED
tests/test_tax_and_statutory_reports.py (3 tests: RBAC 403, slab change + single-month rerun, PF/ESI/TDS) PASSED
tests/test_financial_reports.py (3 tests: Trial Balance 0 mismatches, P&L exact match, BS equation) ..... PASSED
Phase 1 & Phase 2 Regression Suite (96 tests: attendance, payroll 7340/7340, payslips, ledger, AR) ...... PASSED

======================== 113 passed, 0 failed in 192.4s ========================
```

4. What I Learned in Phase 3 (In My Own Words — 4 Lines)

1. **Sub-ledgers and the General Ledger are two views of one truth:** Every vendor bill, customer invoice, and monthly payroll run must post balanced, idempotent journal entries so sub-ledger totals and ledger accounts reconcile to 0.00 every month.
2. **Security must live in the backend data layer, not the UI:** Using bcrypt password hashes, short-lived JWTs, role dependencies, ownership checks, and live is_active == 'Y' database lookups prevents privilege escalation and immediately revokes deactivated users.
3. **Business rules belong in effective-dated tables, not hard-coded constants:** Reading tax slabs and statutory rates from the database lets admins adjust future tax calculations while keeping closed historical payroll runs immutable.
4. **Financial statements require strict hierarchy discipline:** Filtering out parent summary accounts (1000-5000) and closing P&L Net Profit into Equity ensures the Trial Balance, Profit & Loss, and Balance Sheet satisfy Assets = Liabilities + Equity + Profit to the exact paisa.

5. Problems Encountered & How I Solved Them

1. Revoking Access Immediately When a User Is Deactivated While Holding an Active 30-Minute JWT (Task 13)

Problem: Stateless JWT verification only checks cryptographic signature and token expiry (exp), which would allow a terminated employee or revoked admin to keep calling APIs for up to 30 minutes.
Solution: Updated get_current_user() in app/core/security.py to query the users table by sub on every request and verify user.is_active == 'Y', immediately returning 401 USER_DEACTIVATED the moment a user is deactivated.

2. Swagger UI OAuth2 Form Login Compatibility Without External Multipart Dependencies (Task 13)

Problem: Swagger UI's Authorize modal submits credentials as application/x-www-form-urlencoded whereas API clients send JSON, and calling request.form() without python-multipart triggered a 500 error.
Solution: Built a unified parser in POST /api/v1/auth/login using Python's standard library urllib.parse.parse_qs alongside JSON parsing and enabled persistAuthorization: True in FastAPI's Swagger UI configuration.

3. Preventing Double-Counting of Group Accounts and Closing Current Profit into the Balance Sheet (Task 15)

Problem: Summing all Chart of Accounts rows in P&L or Balance Sheet would risk double-counting parent group accounts (1000, 2000, 3000, 4000, 5000), and leaving out unclosed current-period profit would leave Assets higher than Liabilities + Equity by INR 224,683,587.58.
Solution: Explicitly filtered out is_group == 'Y' accounts in report_service.py and computed current_period_profit = Total Revenue - Total Expenses dynamically for the requested as_of date, achieving a 0.00 Balance Sheet variance.

4. Changing Tax Slabs for a Single-Month Rerun Without Corrupting 17 Prior Closed Payroll Runs (Task 14)

Problem: Modifying a tax slab and rerunning payroll could accidentally recalculate or overwrite historical months if all runs were reprocessed globally.
Solution: Scoped POST /api/v1/tax/rerun/{month} to recalculate solely the targeted month (e.g., 2026-09) while leaving PR-2025-04 through PR-2026-08 untouched, then restored the slab and verified all 18 runs reconcile to 0.00.

6. Self-Check & Departmental Profit Reflection (Thinking for Yourself)

Self-Check — Can I solve a new problem in this project without a step-by-step guide?
Yes. Because the system follows a strict three-layer architecture (routes -> services -> models) backed by an immutable double-entry ledger and 113 automated tests, adding a new feature (such as fixed-asset depreciation or multi-regime tax elections) follows a predictable pattern: define the schema/migration, implement pure Decimal service logic that posts balanced journal lines, protect the route with require_role(), and verify with verify_ledgerforge.py.

Departmental Profit Reflection (e.g., Department D01 — Engineering):
In journal_lines, all payroll expenses (5001-5005) and operating bills (5101-5112) carry cost_center_department_id = 'D01', so D01's direct expenses are already known. However, client revenue lines (4001-4010) have cost_center_department_id = NULL because invoices are billed at the company/service-line level, and shared overhead (rent, audit fees) is posted to D00/D09. To show true departmental profit, we would tag invoice lines to delivery cost centers and define an overhead allocation key (e.g., by headcount).

7. Plan for Next Phase (Phase 4 — Production Readiness, Packaging & Release)

• **Containerization & One-Command Startup:** Package the FastAPI application, Alembic migrations, and database seeder into a production Dockerfile and docker-compose.yml with health checks and environment configuration.
• **CI/CD & Quality Gates:** Configure GitHub Actions to run ruff linting, formatting, pytest (113+ tests), and verify_ledgerforge.py on every push and pull request.
• **Performance, Observability & Release Documentation:** Add structured request logging, rate limiting, API documentation polish, and final release notes tagged for the Phase 4 capstone submission.

8. Submission & Git Tag Sign-Off

Candidate Signature: Ancillin P G | **Date:** November 4, 2026
Git Remote URL: https://github.com/Ancilin/Foundation-Project-Setup-Employee-Data-Model
Git Tags Verified: phase-1, phase-2, phase-3, v0.0.1